STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-RINGS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1071-1079 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1081-1107, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_examples-defos.json (pb=1146, stmt 1071-1079, proof 1081-1107).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00RO: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1107–1107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08S7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1107–1107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08LI: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1107–1107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08S5: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1107–1107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DY1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1031–1040. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063Y: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1107–1107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Example 0DY1 let P be a k-algebra. Then*

$$T\mathcal{D}ef_P = Ext^1_P(NL_{P/k}, P) \quad \text{and} \quad Inf(\mathcal{D}ef_P) = Der_k(P, P)$$

Proof. Recall that $\mathrm{Inf}(\mathcal{D}ef_P)$ is the set of automorphisms of the trivial deformation $P[\epsilon] = P \otimes_k k[\epsilon]$ of P to $k[\epsilon]$ equal to the identity modulo ϵ. By Deformation Theory, Lemma 08S5 this is equal to $\mathrm{Hom}_P(\Omega_{P/k}, P)$ which in turn is equal to $\mathrm{Der}_k(P, P)$ by Algebra, Lemma 00RO.

Recall that $T\mathcal{D}ef_P$ is the set of isomorphism classes of flat deformations Q of P to $k[\epsilon]$, more precisely, the set of isomorphism classes of $\mathcal{D}ef_P(k[\epsilon])$. Recall that a $k[\epsilon]$-algebra Q with $Q/\epsilon Q = P$ is flat over $k[\epsilon]$ if and only if

$$0 \to P \xrightarrow{\epsilon} Q \to P \to 0$$

is exact. This is proven in More on Morphisms, Lemma 063Y and more generally in Deformation Theory, Lemma 08LI. Thus we may apply Deformation Theory, Lemma 08S7 to see that the set of isomorphism classes of such deformations is equal to $\mathrm{Ext}^1_P(NL_{P/k}, P)$. □